GDSReady Project Report

RTL-to-GDSII Implementation of a Packet Admission Controller

Project: Packet Admission Controller (PAC)

Technology: Sky130

Flow: RTL → Synthesis → Floorplan → Placement → CTS → Routing → STA/Power → DRC/LVS → GDSII

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1. Abstract and Project Objectives

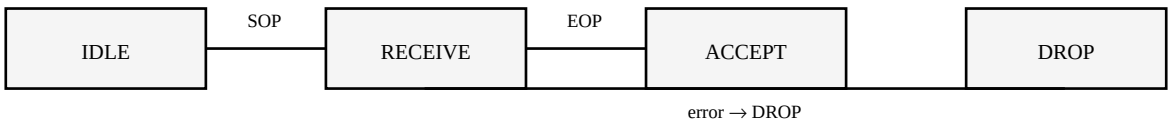
This project demonstrates a complete digital ASIC implementation flow for a finite-state-machine based Packet Admission Controller. The design uses start-of-packet (SOP), end-of-packet (EOP), and error information to control packet reception and admission decisions through in_packet, accept, and drop outputs. The objective is to take synthesizable RTL through an open-source physical-design flow and obtain GDSII while collecting implementation metrics.

The completed run used LibreLane with the Sky130 technology and progressed through synthesis, floorplanning, placement, clock-tree synthesis, routing, post-route timing and power analysis, physical verification, and GDSII stream-out. The run produced multiple GDS representations and a KLayout-rendered layout image. The recorded physical checks include zero KLayout DRC errors, zero Magic DRC errors, zero antenna violations, and zero routing DRC errors.

Objectives

1	Implement a compact packet-admission FSM in synthesizable RTL.
2	Verify functional behavior using a simulation testbench.
3	Synthesize the design to Sky130 standard cells.
4	Perform floorplanning, placement, CTS and routing.
5	Evaluate area, timing and power after physical implementation.
6	Run physical verification and generate GDSII.

Functional concept



Documented state behavior: IDLE → RECEIVE on SOP; RECEIVE → ACCEPT on EOP for a valid packet; RECEIVE → DROP on an error; ACCEPT/DROP → IDLE.

2. Tools, Technology and Implementation Flow

Environment: Ubuntu/WSL with Docker and LibreLane. Successful run directory:

`config/runs/RUN_2026-09-22_15-24-37/`

The successful configuration used a 10 ns clock period (100 MHz target), conservative utilization/density settings, and enabled CTS plus Magic and KLayout stream-out.

Implementation stages

Stage	Purpose
RTL	Packet Admission Controller Verilog
Simulation	Functional verification with testbench
Synthesis	Yosys maps RTL to Sky130 standard cells
Floorplan	Core/die planning and power-grid preparation
Placement	Global and detailed placement
CTS	Clock-tree synthesis
Routing	Global and detailed routing
STA / Power	Post-route timing and power analysis
Physical signoff	DRC, antenna, LVS and stream-out
GDSII	Final layout database and rendered layout

LibreLane configuration used

```
{
  "DESIGN_NAME": "packet_admission_ctrl",
  "VERILOG_FILES": ["dir::../src/packet_admission_ctrl.v"],
  "CLOCK_PORT": "clk",
  "CLOCK_PERIOD": 10,
  "FP_CORE_UTIL": 30,
  "PL_TARGET_DENSITY_PCT": 40,
  "FP_ASPECT_RATIO": 1.0,
  "RUN_CTS": true,
  "RUN_POST_GRT_DESIGN_REPAIR": true,
  "RUN_MAGIC_STREAMOUT": true,
  "RUN_KLAYOUT_STREAMOUT": true
}
```

3. RTL, Source Artifacts and Design Implementation

The project hierarchy contains RTL under src/, verification under tb/, scripts/configuration, and the complete LibreLane run under config/runs/. The recorded run confirms Yosys synthesis, OpenROAD physical implementation, timing/power reporting, DRC stages, an LVS checker stage, and final stream-out artifacts.

Design interface

Port	Direction	Function
clk	Input	System clock
rst	Input	Reset
sop	Input	Start of packet
eop	Input	End of packet
error	Input	Packet error indication
in_packet	Output	Packet reception state
accept	Output	Accepted packet indication
drop	Output	Dropped packet indication

Reference RTL interface listing

```
// Packet Admission Controller — documented interface
// Reference listing only; exact RTL remains in src/packet_admission_ctrl.v
module packet_admission_ctrl (
    input  wire clk,
    input  wire rst,
    input  wire sop,
    input  wire eop,
    input  wire error,
    output wire in_packet,
    output wire accept,
    output wire drop
);
    // FSM behavior:
    // IDLE -> RECEIVE on sop
    // RECEIVE -> ACCEPT on eop when error is inactive
    // RECEIVE -> DROP on error
    // ACCEPT/DROP -> IDLE
endmodule
```

Important: this listing documents the project interface and FSM behavior; it is not presented as a byte-for-byte copy of the original RTL source. The exact source should remain in the accompanying project ZIP under src/packet_admission_ctrl.v.

4. Results and Signoff Metrics

The following values are taken from the completed implementation artifacts and recorded run outputs. Synthesis area, post-implementation physical area, timing, power and verification metrics are kept distinct because they originate from different stages.

Metric	Result
Technology	Sky130
Clock period	10 ns
Target frequency	100 MHz
Synthesis cells	13
Post-P&R standard-cell instances	30
Post-P&R total instances	80
Core area	510.49 µm²
Die area	1567.27 µm²
Utilization	66.18%
Sequential-cell area	85.0816 µm²
Worst setup slack	4.7881 ns
Worst hold slack	0.1147 ns
Setup TNS	0 ns
Hold TNS	0 ns
Setup violations	0
Hold violations	0
Max slew violations	0
Max capacitance violations	0
Power corner	max_ss_100C_1v60
Total power	69.40814 µW
Internal power	58.154 µW
Switching power	11.04609 µW
Leakage power	0.208043 µW
KLayout DRC	0 errors
Magic DRC	0 errors
Antenna violations	0
Routing DRC errors	0
Power-grid violations	0

Area

Yosys synthesis reports 13 mapped cells and a synthesized chip area of 166.4096 in the report's library area units, with 85.0816 attributed to sequential elements (51.13%). The post-implementation instance area is 510.49 µm² and the recorded die area is 1567.27 µm²; these are different-stage quantities.

Timing

The target clock period is 10 ns. Recorded final post-P&R; timing metrics show positive setup and hold slack, zero setup and hold TNS, and zero reported setup/hold violations. A separate nominal-corner metrics record also reports 6.21346 ns setup worst slack and 0.35007 ns hold worst slack; it is retained separately from the max-SS power corner.

Power

At max_ss_100C_1v60, the detailed power report gives 69.40814 µW total: 58.154 µW internal, 11.04609 µW switching and 0.208043 µW leakage. A separate nominal-TT flow metric is approximately 100.1084 µW and should not be substituted for the detailed max-SS result.

5. Physical Verification, Outputs and Submission Structure

The completed run contains final GDSII stream-outs and reports for the digital ASIC flow. KLayout and Magic DRC results record zero errors; antenna violations and routing DRC errors are also zero. An LVS stage is present, but this report does not label LVS PASS without direct confirmation from the final LVS report text.

Key final outputs

Output	Location
Magic GDS	58-magic-streamout/packet_admission_ctrl.gds
KLayout GDS	59-klayout-streamout/packet_admission_ctrl.klayout.gds
Final GDS	final/gds/packet_admission_ctrl.gds
Magic GDS copy	final/mag_gds/packet_admission_ctrl.magic.gds
KLayout GDS copy	final/klayout_gds/packet_admission_ctrl.klayout.gds
Rendered layout	60-klayout-render/packet_admission_ctrl.png
Synthesis statistics	06-yosys-synthesis/reports/stat.rpt
Post-P&R timing summary	56-openroad-stapostpnr/summary.rpt
Detailed power	56-openroad-stapostpnr/max_ss_100C_1v60/power.rpt
KLayout DRC	66-klayout-drc/reports/drc.klayout.json
Magic DRC	65-magic-drc/reports/drc.magic.rpt
LVS report	71-netgen-lvs/reports/lvs.netgen.rpt

Recommended submission package

```
GDSReady_Packet_Admission_Controller.zip
■■■ src/
■■■ tb/
■■■ scripts/
■■■ config/runs/RUN_2026-09-22_15-24-37/
■■■ reports/final/
■   ■■■ synthesis/
■   ■■■ timing/
■   ■■■ power/
■   ■■■ drc/
■   ■■■ lvs/
■■■ final GDS / layout outputs
```

Conclusion

The project demonstrates a complete RTL-to-GDSII implementation of the Packet Admission Controller using Sky130 and LibreLane. The final run produced GDSII outputs and quantitative results for synthesis area, physical area, timing, power and physical verification. The accompanying project ZIP should be treated as the authoritative source package containing the exact RTL, testbench, configuration, complete run directory and generated artifacts.

Before submission, verify that the ZIP contains src/packet_admission_ctrl.v, the testbench, LibreLane configuration, complete run directory, final GDS files, rendered layout image, and collected reports.